

## W3-Ex2 — The Token Bill: Measure It, Then Feel It

Model: `gpt-4.1-mini`. Part 1 reads `prompt_tokens` straight from the API; Part 2 routes both agents through a LiteLLM proxy on `localhost:4000` and reads the meter. Code at `week3/token-bill/`.

### PART 1 — THE REGISTRY TAX

| FIRST LLM CALL OF...                                          | PROMPT_TOKENS |
|---------------------------------------------------------------|---------------|
| Baseline — plain loop (its own 2 wired tools + system prompt) | 158           |
| MCP server mounted and <b>used</b> (1 tool, Pallet Pup)       | 81            |
| MCP mounted, <b>not used</b> (1 tool, capital of France)      | 77            |
| <b>MCP mounted, not used, 6 tools (same question)</b>         | <b>438</b>    |

**Why does the unused server still cost tokens?** Because the registry — every tool's name, description and JSON schema — is serialized into the context of *every* request so the model can decide whether to call it, so it pays rent whether it is used or not: the same question went from 77 to 438 prompt tokens (+72 per tool) just by mounting five more tools. Ten servers like this (~4,400 tokens) in a fifty-turn chat is ~220,000 tokens of pure registry.

### PART 2 — SAME TASK, TWO ROADS

The task: post a comment to the course forum on Àrtemis (the *Agents playground forum*). Road one: the course's minimal CLI agent plus a 190-token `moodle-forum` skill that shows one `curl` call to the Moodle REST API. Road two: OpenCode driving my own Brave browser through Playwright MCP (CDP), same proxy, same model.

| ROAD                                                                    | LLM CALLS | PROMPT         | COMPLETION   | TOTAL TOKENS   |
|-------------------------------------------------------------------------|-----------|----------------|--------------|----------------|
| CLI — <a href="#">“hello from my CLI agent” (d=1568)</a>                | 3         | 1,674          | 169          | 1,843          |
| <b>Browser — <a href="#">“hello from my browser agent” (d=1569)</a></b> | <b>28</b> | <b>362,164</b> | <b>2,489</b> | <b>364,653</b> |

**Ratio: ~198× more expensive through the browser.** The browser road is so costly because every single step — navigate, open the forum, fight the TinyMCE editor, verify — returns a full accessibility snapshot of the page that lands in the context, and the whole growing conversation is re-sent on each of the 28 calls, while the CLI road did the entire job with one `curl` the model already knew how to write. The browser would still be the right choice for a task where no API or CLI exists at all — for example operating a legacy web application (or checking that a page actually *renders* correctly for a human), where the web UI is the only interface there is.